Relationship between Maximum Principle and Dynamic Programming

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1 The saving problem

In this note, we demonstrate the relationship between the continuous-time Maximum Principle (optimal control) and continuous-time Dynamic Programming using a simple saving problem. The problem is

$$\max_{c(t),a(t)} \int_0^\infty e^{-\rho t} u(c(t))dt$$

subject to

$$\dot{a}(t) = ra(t) - c(t),$$

where $c(t)$ is consumption at time t, $a(t)$ denotes asset holdings at time t, $\rho > 0$ is the discount rate, $r > 0$ is the interest rate, and $u(\cdot)$ is an increasing and concave utility function. Here, $\dot{a}(t) = da(t)/dt$ denotes the time derivative of $a(t)$.

2 Maximum Principle

To solve this problem using the Maximum Principle, one can use either of two equivalent formulations. The first method uses the *present-value Hamiltonian*:

$$H^p(t) \equiv e^{-\rho t} u(c(t)) + \lambda(t)(ra(t) - c(t)),$$

where $\lambda(t)$ is the costate variable. The first-order conditions for this formulation are

$$\frac{\partial H^p(t)}{\partial c(t)} = 0$$

and

$$\frac{\partial H^p(t)}{\partial a(t)} + \dot{\lambda}(t) = 0.$$

(The necessary condition for optimality includes the transversality condition, but we omit it here.)

The second method uses the *current-value Hamiltonian*:

$$H^c(t) = u(c(t)) + \mu(t)(ra(t) - c(t)), \tag{1}$$

where $\mu(t)$ is the costate variable in this case. The first-order conditions are

$$\frac{\partial H^c(t)}{\partial c(t)} = 0 \tag{2}$$

and

$$\frac{\partial H^c(t)}{\partial a(t)} + \dot{\mu}(t) - \rho\mu(t) = 0. \tag{3}$$

It is easy to check that both approaches are equivalent, with the relationship $\lambda(t) = e^{-\rho t}\mu(t)$. Therefore, we will work with the current-value Hamiltonian below. For the saving problem above, the first-order condition (2) is

$$u'(c(t)) = \mu(t) \tag{4}$$

and (3) is

$$r\mu(t) + \dot{\mu}(t) - \rho\mu(t) = 0. \tag{5}$$

3 Dynamic Programming

We start by defining the value function. In the stationary formulation, the value function gives the maximized discounted utility from a given initial asset level a:

$$V(a) = \max_{\{c(s)\}_{s\geq 0}} \int_0^\infty e^{-\rho s} u(c(s))\, ds,$$

subject to

$$\dot{a}(s) = ra(s) - c(s), \qquad a(0) = a.$$

Equivalently, starting from time t, we can write

$$V(a(t)) = \max_{\{c(s)\}_{s\geq t}} \int_t^\infty e^{-\rho(s-t)} u(c(s))\, ds,$$

subject to

$$\dot{a}(s) = ra(s) - c(s), \qquad a(t) \text{ given.}$$

To heuristically derive the HJB equation, consider a discrete-time formulation with period length Δ. The discrete-time Bellman equation is

$$V(a(t)) = \max_{c(t), a(t+\Delta)} u(c(t))\Delta + \frac{1}{1+\rho\Delta} V(a(t+\Delta)) \tag{6}$$

subject to

$$a(t+\Delta) = (1 + r\Delta)a(t) - c(t)\Delta. \tag{7}$$

Now, subtract $V(a(t))/(1+\rho\Delta)$ from both sides, which does not affect the maximization. Then take a Taylor approximation of the right-hand side with respect to a around $a(t)$:

$$\frac{\rho\Delta}{1+\rho\Delta}V(a(t)) = \max_{c(t),a(t+\Delta)} u(c(t))\Delta + \frac{1}{1+\rho\Delta}V'(a(t))(a(t+\Delta)-a(t)).$$

Using the constraint and dividing both sides by Δ, we obtain:

$$\frac{\rho}{1+\rho\Delta}V(a(t)) = \max_{c(t)}\ u(c(t)) + \frac{1}{1+\rho\Delta}V'(a(t))(ra(t)-c(t)).$$

Taking the limit as $\Delta \to 0$, the Bellman equation (6) becomes

$$\rho V(a(t)) = \max_{c(t)}\ u(c(t)) + V'(a(t))(ra(t)-c(t)). \tag{8}$$

This equation is the *Hamilton-Jacobi-Bellman (HJB) equation.* Using the continuous-time version of the constraint, that is,[1]

$$\dot{a}(t) = ra(t) - c(t), \tag{9}$$

we can rewrite (8) as

$$\rho V(a(t))dt = \max_{c(t)}\ u(c(t))dt + V'(a(t))\dot{a}(t)dt,$$

which can be interpreted as an "asset equation." That is, the left-hand side is the required return over the time interval dt on an asset with value $V(a(t))$, while the right-hand side is the income gain $u(c(t))dt$ plus the capital gain $V'(a(t))\dot{a}(t)dt$.

The first-order condition of the HJB equation (8) is

$$u'(c(t)) = V'(a(t)). \tag{10}$$

Consider the HJB equation (8) evaluated at the optimal choice of $c(t)$ (that is, $c(t)$ satisfies (10)). For such $c(t)$,

$$\rho V(a(t)) = u(c(t)) + V'(a(t))(ra(t)-c(t))$$

must hold for all $a(t)$, and thus we can take the derivative with respect to $a(t)$ on both sides:

$$rV''(a(t))a(t) + rV'(a(t)) - V''(a(t))c(t) - V'(a(t))\frac{dc(t)}{da(t)} + u'(c(t))\frac{dc(t)}{da(t)} - \rho V'(a(t)) = 0.$$

Rearranging using (9) and (10),

$$rV'(a(t)) + V''(a(t))\dot{a}(t) - \rho V'(a(t)) = 0. \tag{11}$$

[1] This equation can be derived by rewriting the constraint (7) as

$$\frac{a(t+\Delta)-a(t)}{\Delta} = ra(t) - c(t)$$

and taking $\Delta \to 0$.

Now we can see the correspondence between the Maximum Principle equations ((4) and (5)) and the Dynamic Programming equations ((10) and (11)) with $\mu(t) = V'(a(t))$. This relationship also clarifies the interpretation of the Lagrange multiplier $\mu(t)$: it is the marginal value of wealth, measured in current utility units.

Note that, with this relabeling and evaluated at $\mu(t) = V'(a(t))$,

$$\rho V(a(t)) = \max_{c(t)} \; H^c(t),$$

where $H^c(t)$ is the current-value Hamiltonian, defined in (1). The first-order condition (2) is the direct consequence of the right-hand side maximization.